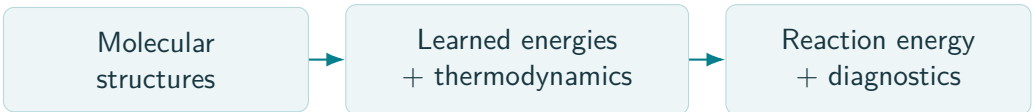
Metabolic reaction thermodynamics

MetaG

From molecular structure to metabolic reaction Gibbs energy



Ray Zhu

October 2026

Why thermodynamics matters for metabolism



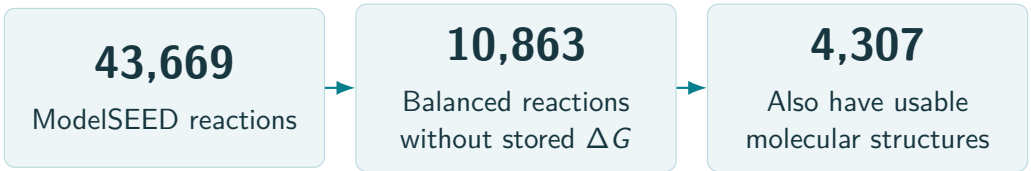
Reaction direction
Which way can it run?

Pathway feasibility
Where is the bottleneck?

Metabolic design
Which route is favorable?

Concentrations and coupling also affect feasibility; a favorable energy does not imply a fast reaction.

The thermodynamic map is incomplete



Molecular structures provide another route to reaction energies.

Local ModelSEED snapshot; live, non-transport reactions. Structural eligibility measures input availability, not accuracy.

Three routes from chemistry to reaction energy

Group methods

Chemical fragments

↓

Fitted group energies

Graph models

Molecular graph

↓

Learned energy estimate

MetaG

3D molecular states

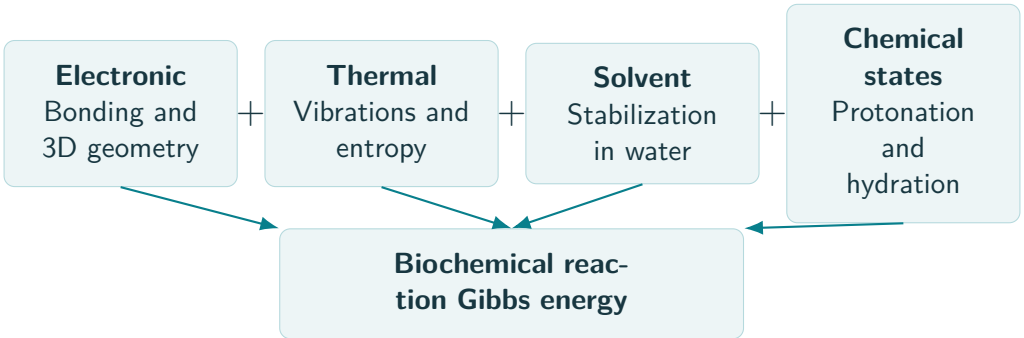
↓

Energy + physical terms

MetaG makes geometry, chemical states, and energy contributions explicit.

Examples: eQuilibrator / dGPredictor; dGbyG. UMA supplies MetaG's learned electronic energies and forces.

A molecular energy is only the starting point



The pipeline connects microscopic chemistry to a biochemical quantity.

UMA is a machine-learned interatomic potential: it predicts energies and forces from 3D structures.

What quantity does MetaG predict?

$$\Delta_r G' = \underbrace{\Delta_r G'^{\circ}}_{\text{MetaG output}} + \underbrace{RT \ln Q'}_{\text{concentration effect}}$$

Standard biochemical conditions

298.15 K | pH 7

$I = 0$ | no Mg binding

Reaction as written

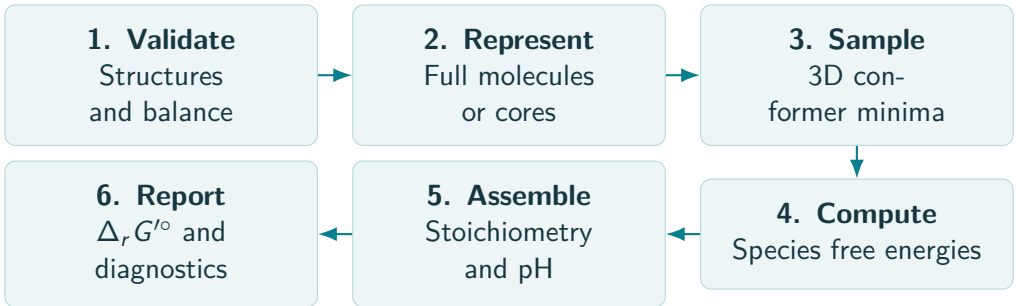
1 M solutes; pure liquid water

Energy reported in kJ/mol

Use the standard energy with concentrations to assess a pathway.

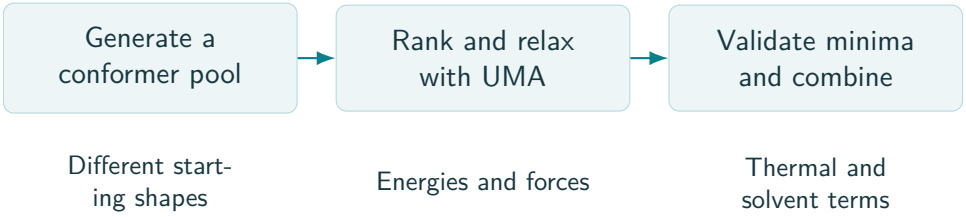
These conditions are fixed in the pipeline. Q' is the biochemical reaction quotient.

The pipeline: structures in, reaction energy out



Each prediction retains the molecular representation and energy terms.

One molecule can occupy several 3D conformations



An ensemble includes both low-energy structures and their populations.

Sampling expands until the calculated ensemble stabilizes. A plateau does not bound undiscovered conformers.

From conformers to species, then to reactions

Within a species

$$g_k = E_k^{\mathrm{UMA}} + G_k^{\mathrm{thermal}} + \Delta G_k^{\mathrm{solvent}}$$

$$G_i = -RT \ln \sum_k e^{-g_k/RT} + C^\circ$$

Low-energy conformers carry more weight.

Across a reaction

$$\Delta_r G'^\circ = \sum_i \nu_i G_i^{\mathrm{eff}} + C_{\mathrm{proton/pH}}$$

Products add; reactants subtract.

G_i^{eff} also includes enabled hydrate states.

Every reaction energy can be traced back to its molecular contributions.

C°: standard-state correction. The appendix gives the proton, water, and pKa conventions.

Large cofactors: retain the chemistry that changes

Current CoA-core representations preserve the thioester

R: identical CoA-derived spectator (charge −4). Teal: preserved thioester.

a rxn02168 · (R)-C4

Input (spectator abbreviated)



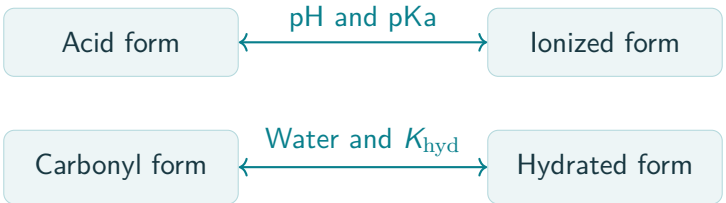
Routed core (scaffold shown explicitly)



The structural core preserves the reactive thioester and its local scaffold.

CoA example from the router. Balance and removed-fragment identity are checked; cores remain approximations.

The aqueous species is a mixture of chemical states



State populations shift the effective free energy of a metabolite.

Production uses independent pKa sites and enabled carbonyl hydration. The two equilibria are schematic; coupled-state routing is not assumed.

Current scoring and the new validation tools

Active reaction scorer

UMA + gas thermal terms + ALPB.

Tabulated pKa patterns; largely independent sites.

Unsupported isotope inputs are withheld; the legacy SMD reaction correction is blocked.

Implemented research tools

Coupled-state registry: references, conditions, and thermodynamic cycles.

Aqueous-ensemble admission and network/covariance audits.

These tools do not replace the default energy or pKa model.

The package now checks more assumptions; its default pKa physics is unchanged.

Integration snapshot: 8 October 2026. Registry transitions require justified state-specific data; the registry does not predict missing pKa values.

Benchmark design separates three questions

Accuracy

364

openTECR reactions

Comparison

302 / 164

Shared / matched targets

Applicability

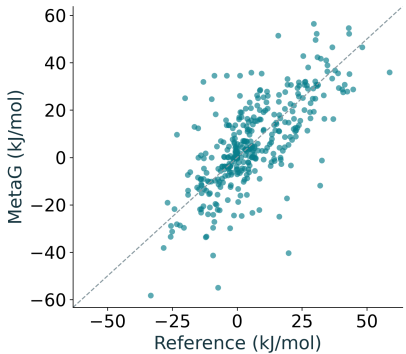
300

Non-TECRDB reactions

Prediction accuracy, relative performance, and coverage are different tests.

Accuracy uses frozen development predictions and archived references; the 300-reaction panel has no experimental energy labels.

openTECR: a 9.54 kJ/mol mean absolute error



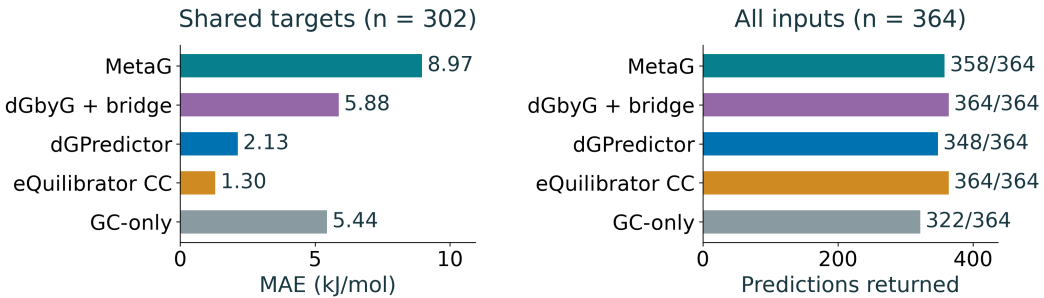
9.54 kJ/mol
Mean absolute error

7.15 kJ/mol
Median absolute error

364 / 364
Reactions scored

Frozen release 2026-10-01c; archived openTECR labels. Development score; reference conditions are not fully resolved.

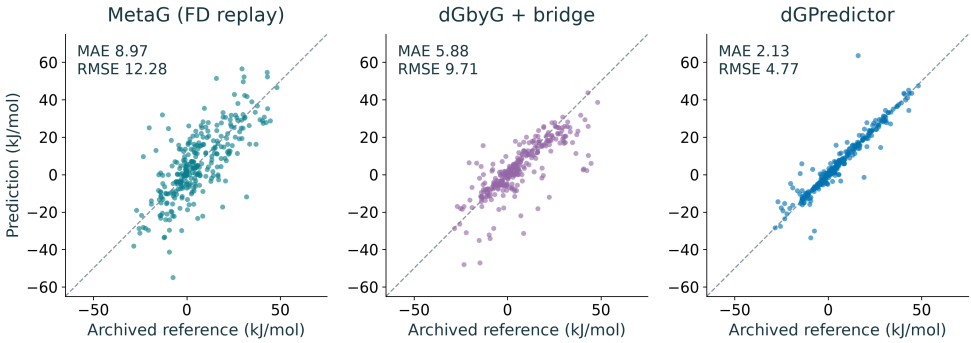
Shared-reaction comparison: accuracy and coverage



Fitted baselines have lower error on this benchmark.

302 shared predictions; pretrained baselines and archived labels. 148 match dGbyG training reactions. MetaG replay withholds six phosphorus cases.

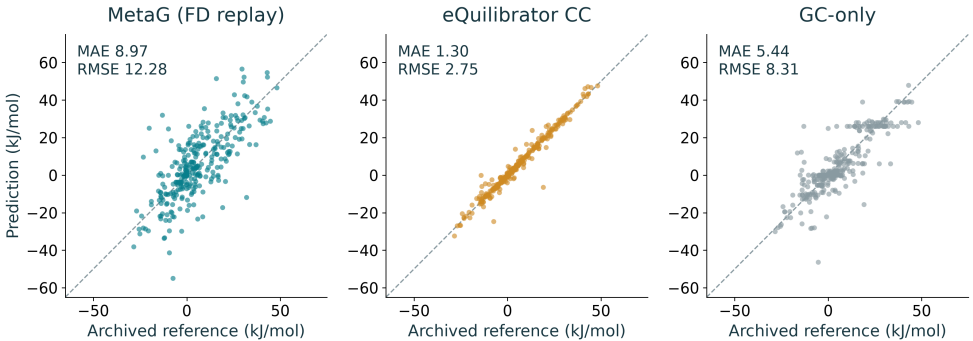
Shared targets: MetaG, dGbyG and dGPredictor



The same 302 reactions and axis limits are used for every method.

Archived labels; MetaG historical FD replay and pretrained baselines. dGbyG uses an eQuilibrator bridge; 148 / 302 targets match its training reactions. This comparison is not held out.

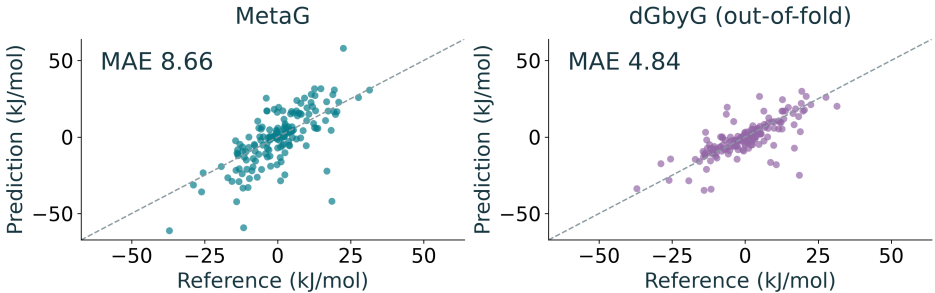
Shared targets: MetaG and eQuilibrator



Component contribution closely reproduces these archived references.

CC: component contribution; GC-only: group contribution. Diagonal: agreement. Archived labels; unresolved conditions and training overlap. This comparison is not held out.

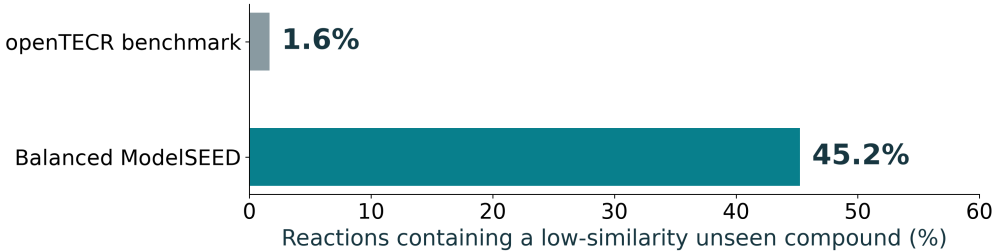
Matched targets: held-out dGbyG is more accurate



MAE: MetaG 8.66 vs dGbyG 4.84 kJ/mol on 164 targets.

Structure, direction, and reaction scale matched. dGbyG is out-of-fold; MetaG is a development estimate bridged to $I = 0.25$ M.

The benchmark covers little of the unfamiliar chemistry



The benchmark contains little of ModelSEED's least familiar chemistry.

Fraction containing an unseen compound with nearest-training Tanimoto < 0.5. Denominators: all balanced ModelSEED reactions vs openTECR. Novelty is not an accuracy measurement.

Why MetaG matters alongside a strong dGbyG

Resolve 3D states

Conformers
Stereochemistry
Aqueous states

Inspect the physics

Separate energy terms
Diagnose sensitivity
Explain model assumptions

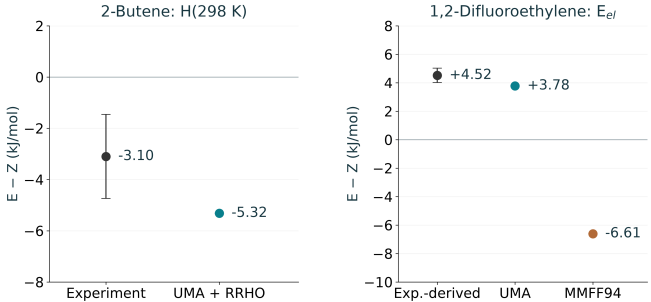
Evaluate new chemistry

Molecular energies
No reaction-label fit
Explicit domain checks

MetaG adds a physical route to estimation and diagnosis.

Complementary capabilities motivate the workflow. Higher accuracy on unfamiliar reactions is not established.

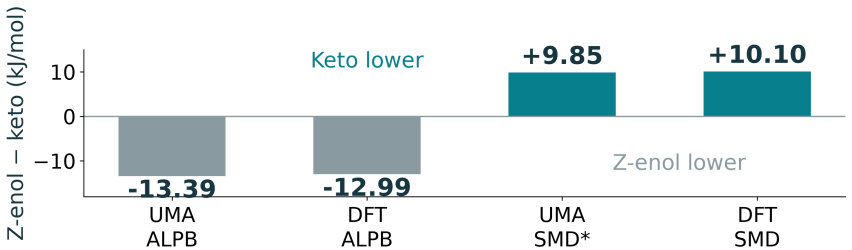
Geometry resolves differences between stereoisomers



UMA reproduces the opposite E/Z orderings of these two gas-phase pairs.

2-butene and 1,2-difluoroethylene. MMFF94 gives the wrong sign for the latter. This component test measures gas-phase energetics.

Energy decomposition identifies solvent sensitivity



Electronic model change
0.41 kJ/mol with ALPB

Solvent model change
23.09 kJ/mol with DFT

Indolepyruvate anions; same five conformers/state and gas thermal terms. ALPB and SMD are continuum models of water. Diagnostic, not full aqueous Gibbs energy. *Solvation term from DFT.

Solvent changes which conformers matter

Same five saved keto conformers

The two source-selected conformers carry:

74.48% under UMA/ALPB

6.50% under DFT/SMD

Source selection misses most target-model weight in this fixed set.

Destination-ensemble audit

Two of three local SMD relaxations meet the force criterion; one reaches the step cap.

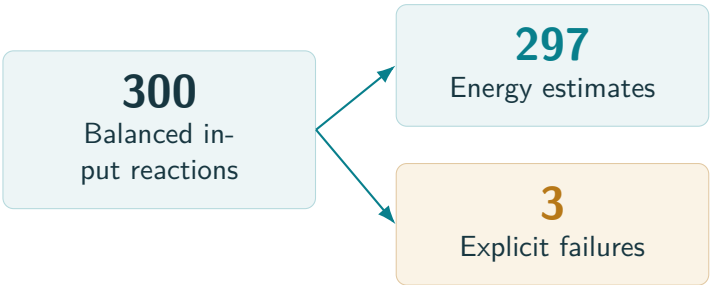
Destination Hessians and thermal terms are absent; starts remain missing.

No aqueous Gibbs ensemble or validated interval is admitted.

Force convergence alone does not validate an aqueous free energy.

Indolepyruvate keto/Z-enol anions. Fixed-set weights retain gas thermal terms; they are diagnostics, not experimental populations or complete aqueous coverage.

ModelSEED: 297 estimates from 300 new reactions



The workflow processes broader chemistry and retains failed cases.

Frozen release; 50 reactions from each of six EC classes, absent from TECRDB. No experimental labels: this measures applicability, not accuracy.

What determines the reliability of an estimate?

Molecular states

Conformer coverage
Protonation
Hydration

Physical models

Continuum solvation
Thermal treatment
Structural cores

Reference match

Temperature and pH
Ionic strength
Mg and phase

Diagnostics make these assumptions visible in each result.

OpenTECR development scores do not establish independent chemical transfer; unsupported phosphorus assignments are withheld.

Network audits expose unresolved consistency

Frozen historical release

49 independent reaction cycles audited.

11 exceed stated rounding bounds.

Context-dependent molecular representations are retained for inspection.

New audit capabilities

Partition by chemical state, conditions, and computational settings.

Preserve failed cases and uncertainty provenance.

Propagate supplied shared-state covariance; none is inferred from scalar intervals.

The audit identifies inconsistency; the reaction energies remain unrepaired.

Historical diagnostic, not a new accuracy evaluation. No independently validated covariance or external interval-coverage result is available.

MetaG: molecular physics for metabolic thermodynamics

The method

- Geometry-resolved species energies
- Physical biochemical corrections
- Traceable estimates and failures

The evidence

- 9.54 kJ/mol development MAE
- 297 / 300 applicability estimates
- Component tests and network audits
- 600 passing software tests

A complementary physical estimator with inspectable chemical assumptions.

MAE and applicability are frozen historical results. Current safeguards and research tools have not established improved overall accuracy or independent transfer.

Appendix: which information is learned or empirical?

Electronic model

UMA is trained on quantum-chemical data. Inputs include geometry, charge, and spin.

Auxiliary properties

pKa references, experimental water hydration, and a component calibration to carbonyl hydration constants.

Reaction labels

openTECR informs evaluation, interval calibration, and method development.

No openTECR reaction-energy offset is fitted in the point estimate.

Benchmark-informed routing makes the reported MetaG accuracy a development score. Class anchor corrections are disabled.

Appendix: implementation validation and scientific scope

Verified implementation

600 passing software tests; package wheel built.

20 protected benchmark artifacts unchanged.

Optional solvent relaxation rejects failed gradients, incomplete starts, and unconverged results; it remains off by default.

Boundaries of that evidence

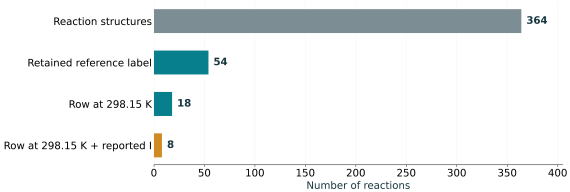
The strict SMD component API has CPU boundary tests; its new wrapper has no fresh GPU numerical validation.

Coupled pKa scoring, complete aqueous thermodynamics, adaptive core corrections, and deployed shared-state covariance remain unresolved.

Software tests do not establish reaction accuracy.

Integration and track sources: `presentation_results.json`. Validation record dated 8 October 2026.

Appendix: reference curation and strict comparison



Condition checks

Free Mg activity; constant definition; measured temperature; ionic strength.

Strict shared subset

Six reactions: MetaG MAE 4.20 kJ/mol. All six overlap dGbyG training.

54 reactions retain curated labels; eight have a row at 298.15 K with reported ionic strength. The small shared subset is descriptive.

Appendix: energy and standard-state conventions

$$\Delta_r G'^{\circ} = \sum_i \nu_i G_i^{\mathrm{eff}} + n_{\mathrm{H}^+} G_{\mathrm{H}^+} + \sum_j s_j m_j \Delta G_j^{\mathrm{pKa}}$$

$$G_i = -RT \ln \sum_k \exp[-(E_k^{\mathrm{UMA}} + G_k^{\mathrm{RRHO}} + \Delta G_k^{\mathrm{ALPB}})/RT] + RT \ln 24.46$$

$$G_i^{\mathrm{eff}} = -RT \ln \left[e^{-G_i/RT} + \sum_S e^{-(G_S - n_S G_{\mathrm{w}})/RT} \right]$$

Thermal terms

Zero-point energy and ideal-gas RRHO; UMA Hessian; 50 cm^{-1} frequency floor; symmetry and linearity.

Reservoir conventions

1 atm $\rightarrow$ 1 M: $+7.93$ kJ/mol.
Pure-water concentration: $+9.96$ kJ/mol.
Proton at pH 7: -1170.8 kJ/mol.

Hydrate terms subtract consumed water. s_j: reaction side; m_j: site multiplicity. Experimental water hydration: -26.4 kJ/mol.

Appendix: benchmark snapshots and comparison regimes

Result	Snapshot / set	Interpretation
Full-panel MAE 9.54	Release 2026-10-01c; 364	Archived labels; development accuracy
Shared-set MAE 8.97	Audited FD replay; 302	Different coverage; pretrained baselines
Matched-set MAE 8.66	164 paper targets	MetaG development vs dGbyG out-of-fold
297 / 300 estimates	Frozen ModelSEED panel	Applicability without energy labels

Later validation changes are not assigned the frozen release's numerical performance.

Condition bridges use eQuilibrator dissociation data. These comparisons do not establish a matched held-out ranking for MetaG.

Appendix: component checks and their scope

Stereoisomer check

Gas-phase 2-butene enthalpy and difluoroethylene electronic-energy differences.

The published dGbyG encoding also collides for tested E/Z and diastereomer inputs.

Solvent component check

Identical saved indolepyruvate geometries; direct ωB97M-V/def2-TZVPD; ALPB vs SMD.

CPU/GPU, grid, and overlap changes are below 0.04 kJ/mol in the tested gaps.

Component evidence explains sensitivity; it does not establish aqueous accuracy.

Gas thermal terms are retained in the solvent comparison. Numerical agreement checks implementation consistency.

Appendix: evidence and reproducibility

Local evidence (relative to the project directory)

Accuracy	`artifacts/results/metag_opentecr_calibrated`
Baseline metrics	`artifacts/benchmark/comparison/full_panel_metrics.tsv`
Matched targets	`artifacts/benchmark/paper_reproduction/metag_vs_dgbyg_pairs.tsv`
ModelSEED applicability	`artifacts/results/generality_report.json`

Database scope and component sources are indexed in `presentation_results.json`.

Fan et al., Cell Systems (2025), doi:10.1016/j.cels.2025.101393. OMol25.
Beber et al., eQuilibrator 3.0, doi:10.1093/nar/gkab1106.
Stereo references: Prosen et al. (1951); Craig et al. (1971), Table II.

Build: make ncs. New presentation figures: make_presentation_figures.py. Scientific figures derive from preserved local evidence.